MIDTERM REVIEW · 26 OCTOBER 2026

Cleany

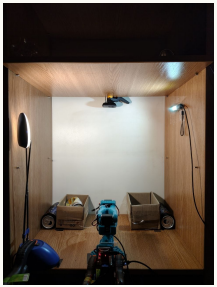
A robot arm that learns to sort trash

Yelyzaveta Bugir · **Nazar Pasichnyk**

github.com/dark-neonus/Cleany_AI_Trash_Sorting_Robot

The task

The arm sees an item and puts it into the right bin





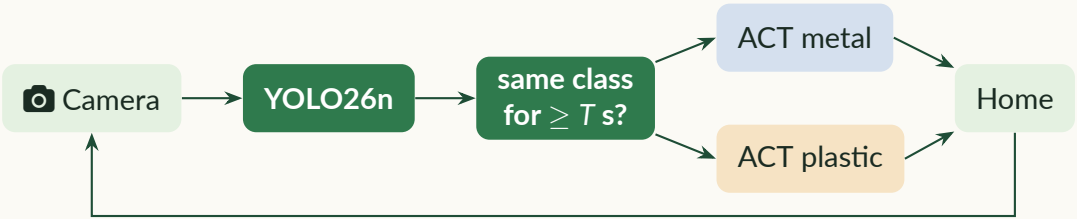
metal → left bin

plastic → right bin

other → ESP32 trap door (planned)

SO-101 arm · front + wrist camera · learned from our teleoperation demos

Current pipeline



$T = 3$ s, confidence ≥ 0.5 · YOLO decides once · one ACT per material · up to 60s per ACT

Why YOLO26n
nano model: learns from 160 frames, runs beside ACT on a 4 GB GPU

Why ACT
learns a task from 50–100 demos; action chunks smooth out noisy teleoperation

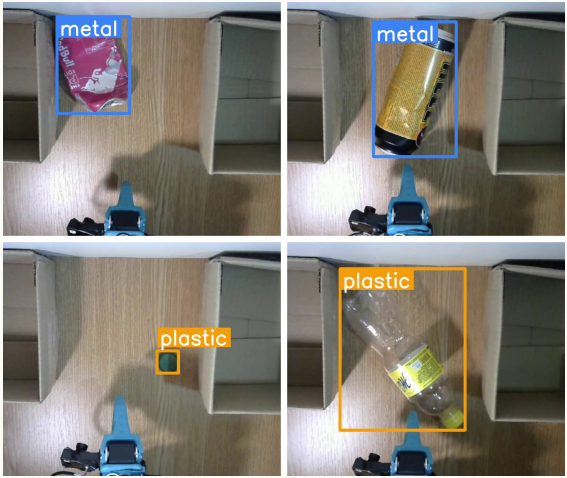
Data

140
demonstrations
60 plastic + 80 metal

80k
frames at 30 fps

160
labelled YOLO frames

all recorded by us
on the Hugging Face Hub



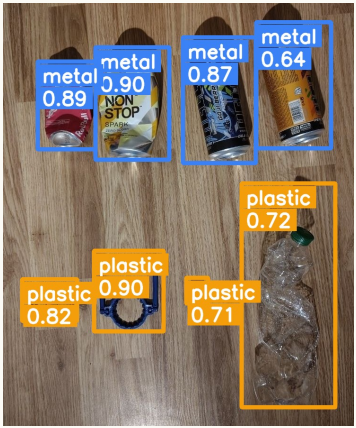
Baseline and metrics

YOLO	val	test
Baseline (Kaggle only)	0.002	0/8
Our frames	**0.870**	**8/8**
+ Kaggle photos	0.869	6/8
Bigger model (26s)	0.858	2 extra

val: mAP_{50-95}, 20 frames · test: unseen phone photo, 8 items

ACT on the arm: plastic 9/10 · metal 8/10

10 trials per model · 100k steps, ∼18 h each on an RTX 2050



our model on the unseen photo

Progress against the proposal

DONE

- test bench and cameras
- 140 demonstrations
- YOLO trained on our data
- 2 ACT models
- end-to-end pipeline
- on the arm: 9/10 and 8/10

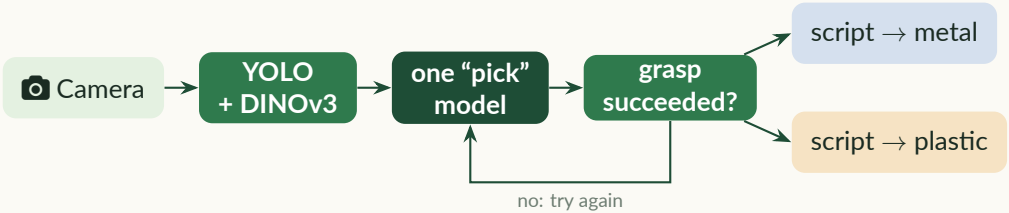
NOT YET · WHY

- “other” class + ESP32 door
  the box is not final yet
- paper and glass
  proposal had 5 classes; 2 end-to-end first
- ACT vs Diffusion Policy
  planned by midterm, moved to November
- SmolVLA
  pilot moved a can, data did not fit the task

Nazar: bench, electronics, cameras, recording + training tools, pipeline visualisation

Yelyzaveta: YOLO (data to evaluation), ACT training, sorting pipeline, repository

New pipeline (being finalised)

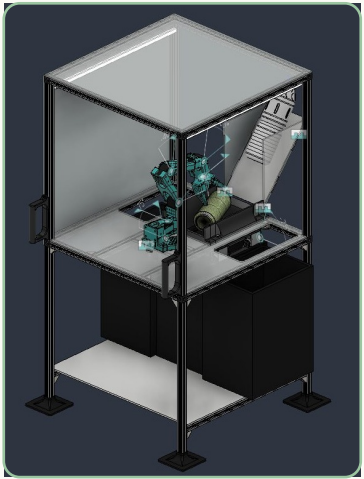


1 model, 2× data
all 140 episodes, cut at the grasp

Placing without ML
fixed poses over the bins
+ open the gripper

Grasp check
gripper servo position + load

Final setup (CAD, real one in progress)



One closed box, the same scene

Fixed lighting
built-in light: no dependence on the room or time of day

Closed environment
frame, walls and doors keep the room out of view

Mixed-trash drop
mechanism for items that are not sorted into a bin

Well-fixed cameras
rigid mounts: the view does not shift between sessions

Why: a model only works in the setup it was recorded in,

so a fixed box makes data, models and results repeatable

Plan, experiments and plan B

NOVEMBER — DECEMBER

“Pick” dataset	ACT vs Diffusion	YOLO vs + DINOv3	Full system
140 episodes + new items	grasp success, 20 trials per model	accuracy on unseen items	30 items in a row: accuracy and time

Baseline: the current two ACTs (9/10, 8/10) · every run, kept or rejected → `results/experiments.md`

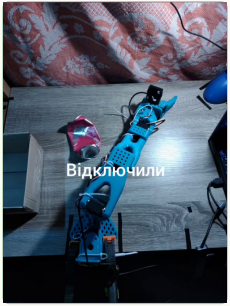
Plan B
one model cannot pick → back to the two ACTs
DINOv3 does not help → YOLO + more labels
no trap door → the arm puts “other” aside

If it works
“other” class via DINOv3 · a pile of items
glass, paper, dirty trash
no laptop: run on a Jetson

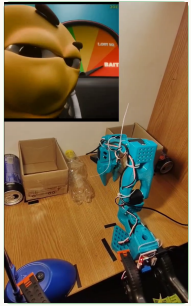
Questions?



“I’m doing my job badly. What could happen?”



…unplugged.



Low IQ or Rage Bait?

Thank you! github.com/dark-neonus/Cleany_AI_Trash_Sorting_Robot